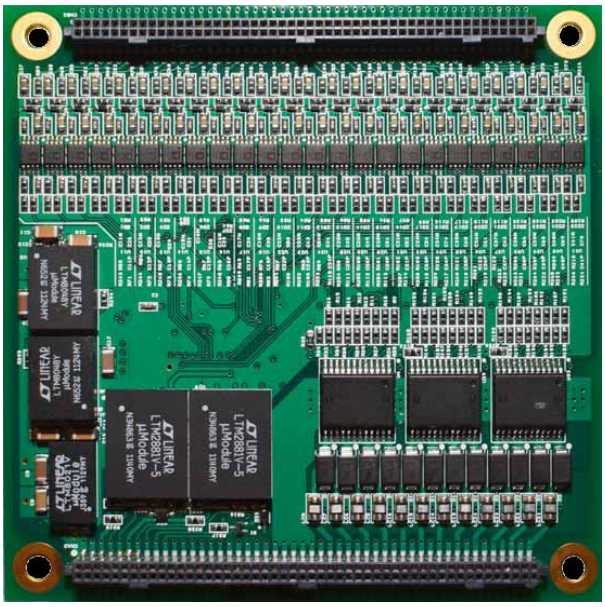
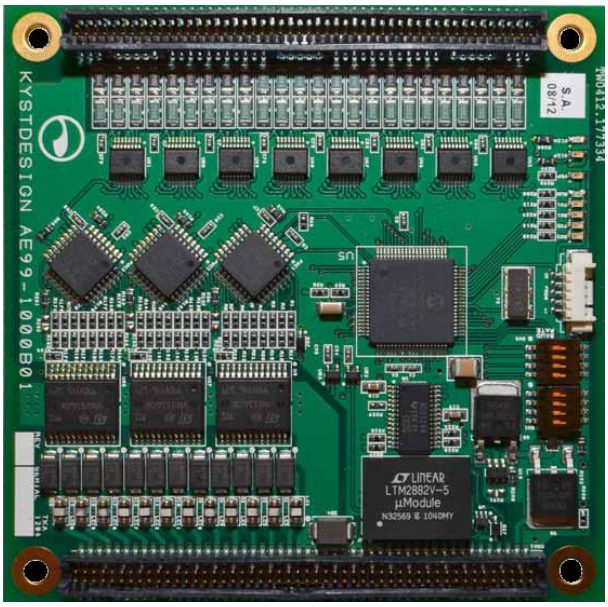
# PROPORTIONAL VALVE DRIVER CARD

## - 24 CH PWM OUTPUTS, 28 CH SENSOR INPUTS

## GENERAL DESCRIPTION

The card is designed to be the backbone of our remote operated systems. It can run 24 proportional valves and read 28 sensors, but is still very compact. The card is based upon years of experience and is made to withstand harsh environment. The board has redundant comms and is stackable to enable redundant operation / expansion.






## FEATURES

- Designed for sub-sea ROV applications
  - Pressure tolerant to 3000 MSW
- Very compact: 85 x 85 mm
- Stackable:
  - Redundant operation
  - Expandable
- Dual Isolated Comms:
  - Redundant link
  - 2 x RS485/RS232
- Supports several telegram formats, with minimum of overhead
- Addressable
- Custom tailored termination boards with LED diagnostics

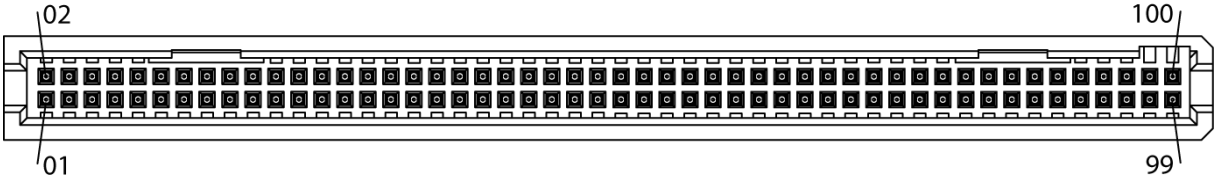
Proportional valve outputs

- 24 x PWM outputs (Proportional valves)
- All outputs well protected
  - short circuit proven
  - current limited

Sensor inputs

- 21 x Analog inputs (0-10vdc / 0-20mA)
  - Individually remotely configurable Volt / current input
- 3 x Digital inputs
- 4 x Temperature Inputs, PT100
  - 24V sensor power
- Diagnostics
  - All sensor inputs can be configured as counter/encoder inputs
  - Transient protection - all inputs
  -

## PINOUTS




| CNA1 & CNA2 | | | |
|---|---|---|---|
| Signal | Pin | | Signal |
| 0Va | 01 | 02 | 0Va |
| +24Va | 03 | 04 | +24Va |
| GND1 | 05 | 06 | +V1 |
| SLO1 | 07 | 08 | TE1 |
| DATA1+ | 09 | 10 | DATA1- |
| GND2 | 11 | 12 | +V2 |
| SLO2 | 13 | 14 | TE2 |
| DATA2+ | 15 | 16 | DATA2- |
| GND3 | 17 | 18 | +V3 |
| DE3 | 19 | 20 | TXD1 |
| RXD1 | 21 | 22 | TXD2 |
| RXD2 | 23 | 24 | RTDX-1 |
| RTD1-2 | 25 | 26 | RTD1-3 |
| RTD2-2 | 27 | 28 | RTD2-3 |
| RTD3-2 | 29 | 30 | RTD3-3 |
| RTD4-2 | 31 | 32 | RTD4-3 |
| 0Vb | 33 | 34 | 0Vb |
| 0Vb | 35 | 36 | 0Vb |
| 0Vb | 37 | 38 | 0Vb |
| 0Vb | 39 | 40 | 0Vb |
| 0Vb | 41 | 42 | 0Vb |
| +24Vb | 43 | 44 | +24Vb |
| +24Vb | 45 | 46 | +24Vb |
| +24Vb | 47 | 48 | +24Vb |
| +24Vb | 49 | 50 | +24Vb |
| +24Vb | 51 | 52 | +24Vb |
| PWM0 | 53 | 54 | PWM0 |
| PWM1 | 55 | 56 | PWM1 |
| PWM2 | 57 | 58 | PWM2 |
| PWM3 | 59 | 60 | PWM3 |
| PWM4 | 61 | 62 | PWM4 |
| PWM5 | 63 | 64 | PWM5 |
| PWM6 | 65 | 66 | PWM6 |
| PWM7 | 67 | 68 | PWM7 |
| PWM8 | 69 | 70 | PWM8 |
| PWM9 | 71 | 72 | PWM9 |
| PWM10 | 73 | 74 | PWM10 |
| PWM11 | 75 | 76 | PWM11 |
| PWM12 | 77 | 78 | PWM12 |
| PWM13 | 79 | 80 | PWM13 |
| PWM14 | 81 | 82 | PWM14 |
| PWM15 | 83 | 84 | PWM15 |
| PWM16 | 85 | 86 | PWM16 |
| PWM17 | 87 | 88 | PWM17 |
| PWM18 | 89 | 90 | PWM18 |
| PWM19 | 91 | 92 | PWM19 |
| PWM20 | 93 | 94 | PWM20 |
| PWM21 | 95 | 96 | PWM21 |
| PWM22 | 97 | 98 | PWM22 |
| PWM23 | 99 | 100 | PWM23 |

| CNB1 & CNB2 (1) | | | |
|---|---|---|---|
| Signal | Pin | | Signal |
| +5V | 01 | 02 | 0V |
| IN (OUT) (1) | 03 | 04 | AIN0 |
| OUT (IN) (1) | 05 | 06 | SPW0 |
| | 07 | 08 | AIN1 |
| | 09 | 10 | SPW1 |
| | 11 | 12 | AIN2 |
| | 13 | 14 | SPW2 |
| | 15 | 16 | AIN3 |
| | 17 | 18 | SPW3 |
| | 19 | 20 | AIN4 |
| | 21 | 22 | SPW4 |
| | 23 | 24 | AIN5 |
| | 25 | 26 | SPW5 |
| | 27 | 28 | AIN6 |
| | 29 | 30 | SPW6 |
| | 31 | 32 | AIN7 |
| | 33 | 34 | SPW7 |
| | 35 | 36 | AIN8 |
| | 37 | 38 | SPW8 |
| | 39 | 40 | AIN9 |
| | 41 | 42 | SPW9 |
| | 43 | 44 | AIN10 |
| | 45 | 46 | SPW10 |
| | 47 | 48 | AIN11 |
| | 49 | 50 | SPW11 |
| | 51 | 52 | AIN12 |
| | 53 | 54 | SPW12 |
| | 55 | 56 | AIN13 |
| | 57 | 58 | SPW13 |
| | 59 | 60 | AIN14 |
| | 61 | 62 | SPW14 |
| | 63 | 64 | AIN15 |
| | 65 | 66 | SPW15 |
| | 67 | 68 | AIN16 |
| | 69 | 70 | SPW16 |
| | 71 | 72 | AIN17 |
| | 73 | 74 | SPW17 |
| | 75 | 76 | AIN18 |
| | 77 | 78 | SPW18 |
| | 79 | 80 | AIN19 |
| | 81 | 82 | SPW19 |
| | 83 | 84 | AIN20 |
| | 85 | 86 | SPW20 |
| | 87 | 88 | DIN21 |
| | 89 | 90 | SPW21 |
| | 91 | 92 | DIN22 |
| | 93 | 94 | SPW22 |
| | 95 | 96 | DIN23 |
| | 97 | 98 | SPW23 |
| | 99 | 100 | 0Va |

Note 1: CNB2 pin numbers 03 and 05 shown in (brackets).

KYSTDESIGN
SUB-SEA TECHNOLOGY

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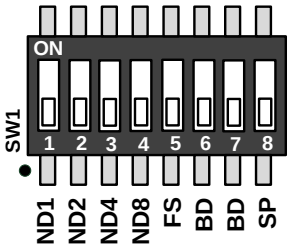
# DIP SWITCHES

## DIP SW1 – SERIAL COMMUNICATION SETTINGS




| SW | NAME | DESCRIPTION |
|---|---|---|
| 1 | N1 | NODE ADDRESS |
| 2 | N2 | |
| 3 | N3 | |
| 4 | N4 | |
| 5 | FS | FAIL SAFE |
| 6 | BD | BAUDRATE |
| 7 | BD | |
| 8 | SP | SPARE (not used) |

### SW1.1-4 NODE ADDRESS

| N1 | N2 | N3 | N4 | NODE |
|---|---|---|---|---|
| OFF | OFF | OFF | OFF | 0 |
| ON | OFF | OFF | OFF | 1 |
| OFF | ON | OFF | OFF | 2 |
| ON | ON | OFF | OFF | 3 |
| OFF | OFF | ON | OFF | 4 |
| ON | OFF | ON | OFF | 5 |
| OFF | ON | ON | OFF | 6 |
| ON | ON | ON | OFF | 7 |
| OFF | OFF | OFF | ON | 8 |
| ON | OFF | OFF | ON | 9 |
| OFF | ON | OFF | ON | 10 |
| ON | ON | OFF | ON | 11 |
| OFF | OFF | ON | ON | 12 |
| ON | OFF | ON | ON | 13 |
| OFF | ON | ON | ON | 14 |
| ON | ON | ON | ON | 15 |

### SW1.5 FAIL SAFE

| FS | DESRIPTION |
|---|---|
| OFF | All outputs will be switched off |
| ON | All outputs will retain previous state |
| This switch controls board's outputs in case of loss of communication. | |

### SW1.6-7 BAUDRATE

| BD6 | BD7 | BAUD RATE |
|---|---|---|
| OFF | OFF | 9600 |
| ON | OFF | 38400 |
| OFF | ON | 57600 |
| ON | ON | 115200 |

### SW1.8 SPARE (NOT USED)

# LED INDICATORS

## POWERSUPPLY

| LED | INDICATOR |
|---|---|
| V5P | +5V LOGIC SUPPLY |
| L12VP | +12V ANALOG SUPPLY |
| L12VN | -12V ANALOG SUPPLY |

## COMMUNICATION

| LED | INDICATOR |
|---|---|
| TX1 | SERIAL TRANSMIT |
| RX1 | SERIAL RECEIVE |
| TX1 | SERIAL TRANSMIT |
| RX1 | SERIAL RECEIVE |

## DIAGNOSTIC

| LED | INDICATOR |
|---|---|
| V3 | COMMUNICATION |
| COMMUNICATION LOSS IS INDICATED AFTER 5 SECONDS BY BLINKING LRUNU6 | |

# SPECIFICATIONS

## ELECTRICAL

### POWER SUPPLY

Input Voltage: ..... 24VDC (19 to 36VDC)

Power Requirement: ..... Load Dependent

Internal Power Dissipation: ..... 0.1W

### COM INTERFACE

Standard Interface ..... RS-232/485

Baud Rate..... 9600, 38400, 57600 and 115200

Protocol support..... D-Bus and D2-Bus

### ANALOG INPUTS

Analog Input type: ..... Voltage or Current

Resolution ..... 11 bit

Voltage Range: ..... 0 to 10VDC

Current Range: ..... 0 to 20mA

Switchable Current Shunt ..... 449R

Sensor Power supply ..... 24V

Short Circuit Protection ..... 0.6A

### ANALOG OUTPUTS

Voltage Output level: ..... 0 to 10VDC

### PWM OUTPUTS

Voltage Output level: ..... 24VDC (19 to 36VDC), same as supply voltage

Short Circuit Protection ..... 3.8A

PWM frequency ..... 279Hz

## MECHANICAL

Dimensions: ..... 248.92 (L) x 90.81 (W) x 17.90 (H)

## ENVIRONMENTAL

Pressure Tolerant

Pressure tested to: .....3500meter (350bar)

Temperature: ..... 0° C to +50° C operating

-20° C to +60° C storage

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